

Spherical Pressure Vessel Thickness Calculation

Problem Statement

F1_4: Calculate the required thickness of the shell of a spherical pressure vessel with a diameter of 450 mm. The vessel must withstand an internal fluid pressure of 7 MPa without exceeding a material stress of 70 MPa. Additionally, if the shell is constructed by bolting together two flanged halves using sixteen bolts, determine the root area of each bolt. The tensile stress in the bolts must not exceed 150 MPa.

Given Data

- **Diameter:** 450 mm
- **Internal Pressure:** 7 MPa
- **Material Stress Limit:** 70 MPa

1. Shell Thickness Calculation

For a **thin-walled spherical pressure vessel**, the hoop stress equation is:

$$\sigma = \frac{pd}{4t}$$

Rearranging for t :

$$t = \frac{pd}{4\sigma}$$

where:

- $p = 7 \text{ MPa} = 7 \times 10^6 \text{ Pa}$ (internal pressure),
- $d = 450 \text{ mm} = 0.45 \text{ m}$ (diameter of the vessel),
- $\sigma = 70 \text{ MPa} = 70 \times 10^6 \text{ Pa}$ (allowable stress).

Substituting the values:

$$t = \frac{(7 \times 10^6) \times 0.45}{4 \times (70 \times 10^6)}$$

$$t = \frac{3.15 \times 10^6}{2.8 \times 10^8}$$

$$t = 0.01125 \text{ m} = \boxed{11.25 \text{ mm}} \text{ 4 marks}$$

Thus, the required shell thickness is **11.25 mm**.

2. Bolt Root Area Calculation

The force exerted by the internal pressure is:

$$F = pA$$

where:

$$A = \frac{\pi}{4}d^2$$

Substituting $d = 0.45 \text{ m}$:

$$A = \frac{\pi}{4}(0.45)^2$$

$$A = \frac{\pi}{4} \times 0.2025$$

$$A = 0.159 \text{ m}^2$$

Now, the total force due to pressure:

$$F = (7 \times 10^6) \times 0.159$$

$$F = \boxed{1.113 \times 10^6 \text{ N}} \text{ 3 marks}$$

Since **sixteen bolts** carry this force, the force per bolt is:

$$F_b = \frac{F}{16} = \frac{1.113 \times 10^6}{16}$$

$$F_b = 69,562.5 \text{ N}$$

The required root area of each bolt is:

$$A_b = \frac{F_b}{\sigma_b}$$

where:

- $\sigma_b = 150 \text{ MPa} = 150 \times 10^6 \text{ Pa}$.

$$A_b = \frac{69,562.5}{150 \times 10^6}$$

$$A_b = 4.6375 \times 10^{-4} \text{ m}^2 = \boxed{463.75 \text{ mm}^2} \text{ 3 marks}$$

Thus, the required root area of each bolt is **463.75 mm²**.